

Ultra-Condensed Time Series Formula Sheet

AR(p): $x_t = c + \sum \phi_i x_{t-i} + w_t$ MA(q): $x_t = \mu + w_t + \sum \beta_i w_{t-i}$ ARIMA(p,d,q): Difference d times, then ARMA(p,q) SARIMA(p,d,q)(P,D,Q)m Differencing: Regular: $x_t - x_{t-1}$ Seasonal: $x_t - x_{t-m}$
Identification: AR $\rightarrow$ PACF cuts off MA $\rightarrow$ ACF cuts off ARMA $\rightarrow$ both tail Ljung-Box: want p > 0.05
AIC/BIC: lower better EWMA: $EMAt = \alpha x_t + (1-\alpha)EMAt-1$ Forecast: $x_{t+1} = EMAt$